

Constrained sphere packing and the interpolation problem for positive definite functions

James Eldred Pascoe

Drexel University

Joint work with Sujit Sakharam Damase, IISc, Bangalore

A photograph of two young men standing in front of a large chalkboard in a lecture hall. The man on the left is wearing a green t-shirt with a graphic and a black jacket, while the man on the right is wearing a light-colored patterned sweater. The chalkboard is filled with handwritten mathematical equations and diagrams. The left side of the board contains various mathematical expressions, including $X = \{x_i\}^n$, $G = \mathbb{S}_n \times (\mathbb{Z}/2\mathbb{Z})$, and $I(X) = \{1 - x_i\}$. The right side features a diagram of a graph G with vertices v_1, v_2, v_3, v_4 and edges, along with equations like $y = \bigoplus_{i=1}^n x_i$ and $\frac{1}{2} \sum_{i=1}^n x_i^2 = 10/12$. The background shows a window with a view of a building exterior.

2 / 29

Acknowledgments

We would like to thank:

- The Indian Institute of Science, Bangalore
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Permanently multiplying the network in all areas with an understanding of the transient nature of division is key to a continuing human role in mathematical progress.

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Permanently multiplying the network in all areas with an understanding of the transient nature of division is key to a continuing human role in mathematical progress. “Proof and conjecture, and keep the SF’s score low.” Paul Erdős

Sphere packing

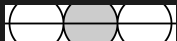
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Want to find the the optimal packing of spheres in some dimension.

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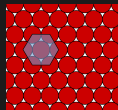
Want to find the optimal packing of spheres in some dimension.

$$n = 1$$



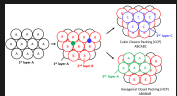
$$n = 2$$

Lagrange



$$n = 3$$

Hales (90s)



$$n = 8, 24$$

Viazovska et al. (2010s)

$$E_8,$$

Leech lattice

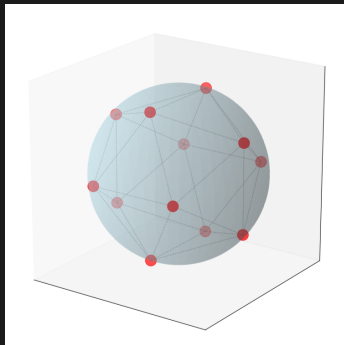
Spherical codes

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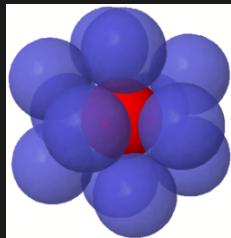
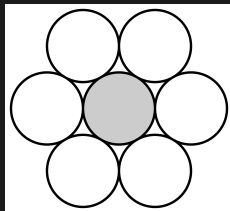
Kissing number problem

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The kissing number is equal to the maximal size of a spherical code with minimum angle $\pi/3$.

Delsarte's method

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Observation: Let $\pi/2 < \theta < \pi$. Any spherical code with minimum angle θ has cardinality less than

$$\frac{\cos \theta - 1}{\cos \theta}.$$

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Strategy: Given $n \in \mathbb{N}$ and $0 < \theta < \pi/2$ find the maximum $\tilde{\theta} > \pi/2$ such that there is a map from the sphere in n -dimensional space to the sphere in a larger dimensional space such that every spherical code with minimum angle θ is mapped to a code with minimum angle at least $\tilde{\theta}$.

Positive definite functions

Positive definite functions

Let G be group, let H be subgroup of G . A (completely) positive definite function

$$f : G/H \rightarrow \mathbb{R} \text{ (resp. self-adjoint operators)}$$

satisfies: $[f(g_i^* g_j)]_{i,j}$ is positive semidefinite for all $g_1, \dots, g_d \in G$.

Stinespring factorization

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Given a positive definite function $f : G/H \rightarrow \mathbb{R}$ it has a **Stinespring factorization**

$$f(g) = v^* \pi(g) v$$

where π is a representation of G and v is a vector which is invariant for $\pi|_H$. If f is continuous and G is a compact group, the Peter-Weyl decomposition gives this is a direct sum of irreducibles. (convex combination)

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The homomorphic analogue of **Tomiyama's theorem** gives that f is in fact bi-invariant,

$$f(h_1 g h_2) = f(g), \forall h_1, h_2 \in H, g \in G.$$

Schoenberg's theorem on positive definite functions

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Let $G = SO(n)$, $H = 1 \oplus SO(n-1)$, $S^{n-1} \cong G/H$. By bi-invariance we have that $f(\gamma)$ depends only on the $1-1$ entry of γ .

Schoenberg's theorem on positive definite functions

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If f is positive definite and continuous, $f(g) = \sum_{i=1}^{\infty} a_i C_i^{(n)}(\gamma_{11})$ where $a_i \geq 0$ and the $C_i^{(n)}$ are Gegenbauer polynomials, a natural basis for bi-invariant functions. (We will normalize to have $C_i^{(n)}(1) = 1$) (Schoenberg '42, Bochner '41)

Operations on positive definite functions

Operations on positive definite functions

Given $C_i^{(n)}(\gamma_{11}) = v_i^* \pi_i(\gamma) v_i$, we see that

$$C_i^{(n)} C_j^{(n)} = \sum_{k=0}^{i+j} a_k C_k^{(n)}$$

where $\sum a_k = 1, a_k \geq 0$ as

$$C_i^{(n)}(\gamma_{11}) C_j^{(n)}(\gamma_{11}) = (v_i \otimes v_j)^* (\pi_i \otimes \pi_j)(\gamma) (v_i \otimes v_j)^*$$

and $\pi_i \otimes \pi_j$ may again be broken into irreducibles.

Kernel embedding

Kernel embedding

Theorem (Aronszajn's theorem)

Let X be a set. Let $A : X \times X \rightarrow \mathbb{C}$ such that $[A(x_i, x_j)]_{i,j}$ is positive semidefinite for arbitrary finite collections $x_1, \dots, x_d$.

There exists a Hilbert space $\mathcal{H}$ and $\iota_x : X \rightarrow \mathcal{H}$ such that $\langle \iota_x, \iota_y \rangle = A(x, y)$.

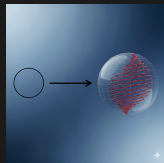
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Thus a positive definite function $f : [-1, 1] \rightarrow \mathbb{R}$ such that $f(1) = 1$ induces an embedding via the kernel $f(\langle x, y \rangle)$ which lands in the sphere of the Hilbert space as $f(\langle x, x \rangle) = 1$.



Delsarte's method

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Theorem (Delsarte '72)

Let $0 < \theta < \pi$. Let $g : [-1, 1] \rightarrow \mathbb{R}$ be a positive definite function on the sphere in n dimensions such that $g(1) = 1$ and $g(x) \leq 0$ if $x \leq \cos \theta$.

Let $\bar{g}$ the constant coefficient of g in its Gegenbauer expansion. (That is the average value of $g(\langle x, y \rangle)$ over x, y in the unit sphere. Let $\mathcal{C}$ be spherical code with minimal angle θ . Then,

$$|\mathcal{C}| \leq \frac{1}{\bar{g}}.$$

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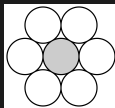
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Linear programming methods: Delsarte-Goethals-Seidel, '77 Kabitsky-Levenshtein '78.

Kissing number in $\mathbb{R}^2$

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The polynomial

$$p(t) = \frac{4}{9}(t - 1/2)(t + 1/2)^2(1 + t)$$

satisfies

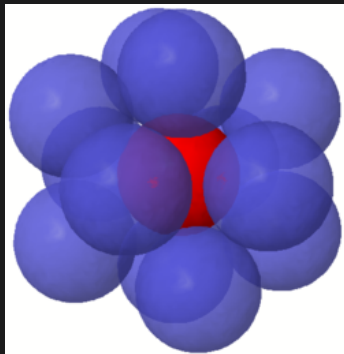
$$p(x) = \frac{1}{6}C_0^{(2)}(x) + \frac{1}{3}C_1^{(2)}(x) + \frac{5}{18}C_2^{(2)}(x) + \frac{1}{6}C_3^{(2)}(x) + \frac{1}{18}C_4^{(2)}(x)$$

$$\text{i.e. } p(\cos \theta) = \frac{1}{6} + \frac{1}{3} \cos \theta + \frac{5}{18} \cos 2\theta + \frac{1}{6} \cos 3\theta + \frac{1}{18} \cos 4\theta$$

Open Problem: What do these coefficients mean?

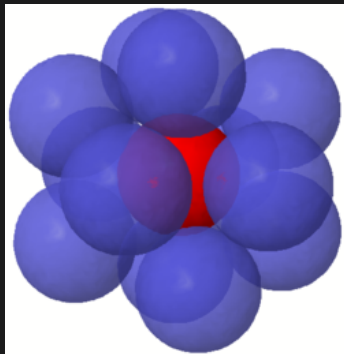
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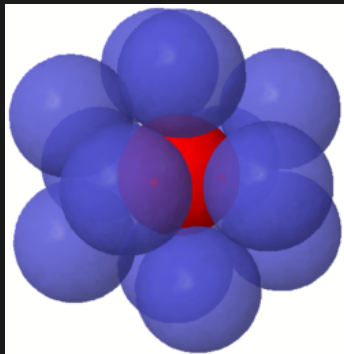
The kissing number is 12, but Delsarte gives an estimate greater than 13.

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Kissing number in $\mathbb{R}^3$



The kissing number is 12, but Delsarte gives an estimate greater than 13. Packing is loose. Musin resolves using two kernel estimates. ('06)

Delsarte sharpness

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where $\bar{g}$ is the spherical average of g .

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Sharp

$n = 2$ kissing number

$n = 8, 24$ kissing number

Sloane, Odlyzko, 79

Not Sharp

$n = 3, 4$ kissing number

Musin

$X = \{-1, 1/2, -1/2, 1\}, n = 10$

Damase, P. ($\frac{1}{\bar{g}} = 46.$)

Cohn-Elkies method

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Theorem (Cohn-Elkies, '03)

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is an admissible function, is not identically zero, and satisfies the following two conditions:

- 1 $f(x) \leq 0$ for $|x| \geq 1$, and
- 2 $\hat{f}(t) \geq 0$ for all t .

Then the center density of n -dimensional sphere packings is bounded above by

$$\frac{f(0)}{2^n \hat{f}(0)}.$$

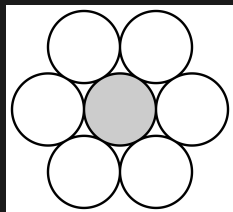
Constrained angle codes

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For example, the spherical code given by six equispaced points on the circle has cosines in $\{0, -1, 1/2, -1/2\}$.

Interpolation for positive definite functions

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c.f. Krein, Jorgensen, etc. on similar but different interpolation problems. ($\Gamma = \Gamma^{-1}$, not necessarily symmetric, only ask for f to induce a kernel on Γ .)

Arveson extension and pathological Class I representations

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Say G/H is **pathological** if there exists a positive definite function on G/H which is not a pointwise limit of continuous ones.

Theorem (Damase-P.)

$SO(\infty)/1 \oplus SO(\infty)$ and $SO(2)/\{1\}$ are not pathological.

Open Problem: $SO(n)/SO(n-1)$? Open Problem: Pathological Delsarte estimates arising from nonmeasurable pd functions.

Continuous extension problem

Continuous extension problem

Let X be a finite subset of $[-1, 1]$. Let $f : X \rightarrow \mathbb{R}$. When is there a continuous positive definite function on the sphere in n -dimensions $\tilde{f} : [-1, 1] \rightarrow \mathbb{R}$ such that $\tilde{f}|_X = f$?

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Theorem (Damase, P.)

Let f correspond to a solvable continuous extension problem. There is a polynomial solution $\tilde{f}$.

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Viazovska et al. functions for Cohn-Elkies are modular functions. (Low algebraic complexity. Open problem: Uniqueness.)

Delsarte hallucinations

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- Consider $X = \{-1, 1/2, -1/2, 1\}$, $n = 10$. There is no corresponding constrained angle code, but we have $\frac{1}{g} = 46$ is an integer. Many such examples exist of **Delsarte hallucinations**.

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Delsarte hallucinations

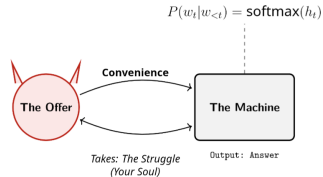
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- On the other hand, the example $X = \{1/3, -1/3, 1\}$ has a spherical code with 10 points in the 5 dimensional sphere, but the sign pattern comes from a Petersen graph.
- Given an X , determining if there exists a subset of the n -sphere with cosines in X of some size k is a difficult combinatorial matrix completion problem.

The future is now

Algebra is the offer made by the devil to the mathematician. The devil says: I will give you this powerful machine, it will answer any question you like. All you need to do is give me your soul: give up geometry and you will have this marvelous machine.



—Sir Michael Atiyah, 2002

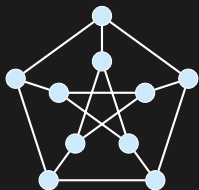


AI is the offer made by the devil to the mind.

The devil says: I will give you this powerful machine, it will complete any thought you like.

All you need to do is give me your soul: give up **the struggle** and you will have this marvelous machine.

— Gemini, 2025



The Petersen Graph

Seidel Matrix S

$$\begin{pmatrix} 0 & - & - & 1 & 1 & - & 1 & 1 & 1 & 1 \\ - & 0 & - & - & 1 & 1 & - & 1 & 1 & 1 \\ - & - & 0 & - & - & 1 & 1 & - & 1 & 1 \\ 1 & - & - & 0 & - & 1 & 1 & 1 & - & 1 \\ 1 & 1 & - & - & 0 & 1 & 1 & 1 & 1 & - \\ - & 1 & 1 & 1 & 1 & 0 & 1 & - & 1 & - \\ 1 & - & 1 & 1 & 1 & 1 & 0 & 1 & - & - \\ 1 & 1 & - & 1 & 1 & - & 1 & 0 & - & - \\ 1 & 1 & 1 & - & 1 & 1 & - & - & 0 & - \\ 1 & 1 & 1 & 1 & - & - & - & - & - & 0 \end{pmatrix}$$

Spectrum: $3^{(5)}, -3^{(5)}$

Gram Matrix G

Construct $X = \{1, \pm 1/3\}$:

$$G = I + \frac{1}{3}S \succeq 0$$

Discovery: $AI = \text{愛}$

- Example suggested by **Gemini**.
- Verified & Formalized by **Damase & P.**

Soft Thresholding

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Theorem (Damase, P.)

The faithfulness constant of $[-\varepsilon, \varepsilon]$ does not go to 1 if $N \geq 4$.

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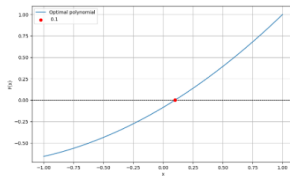


FIGURE 1. The thresholding function for $K = \{0.1\}$, $n = 3$

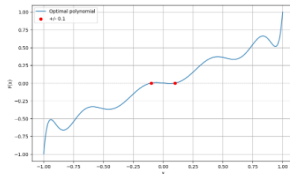


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